

Chapter 02: Biological Neurons → Artificial Neurons

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Chapter Introduction

Once we understand why deep learning emerged, it's time to study its fundamental building block.

A simple biological idea inspired one of the most important mathematical models in modern AI.

Conception of Artificial Neuron and Neural Network

Where did the idea of a neural network come from?

Did scientists literally copy the human brain? Not exactly.

They borrowed a simple but powerful idea from the brain: **Intelligence may emerge from many simple processing units working together.**

This inspired us to set out for a simple unit, so much that no lower bound of allowed simplicity was defined.

The Human Brain

The human brain contains roughly 86 billion neurons.

Each neuron is individually simple, compared to the brain functions that neurons together produce:

- Vision
- Memory
- Language
- Creativity
- Reasoning
- Movement

This observation inspired early AI researchers: **Could intelligence emerge from many simple computational units connected together?**

A Biological Neuron

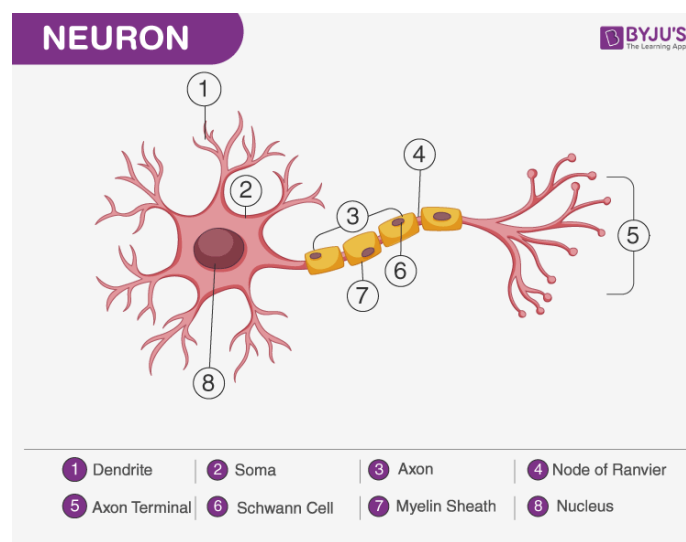


Image Credit: [Byjus](#)

A simplified biological neuron consists of four main parts.

Dendrites (1 in Figure above): Receive signals from other neurons.

Soma (Cell Body, 2 in Figure above): Processes incoming signals. If enough stimulation is received, the neuron "fires."

Axon (3 in Figure above): Carries the output signal.

Synapse (5 in Figure above): The connection between neurons. Signals pass across synapses to other neurons.

Function of a Biological Neuron

Very roughly,

1. Receive Input Signals
2. Combine Input Signals
3. If strong enough, Fire, that is, Send Signal to Next Neurons

Notice something important: **The neuron is not constantly active. It decides whether to send a signal.** Artificial Neurons adopted this behavior too.

The Artificial Neuron

While defining Artificial Neuron, researchers dramatically simplified biology, dropped chemistry and electricity, and kept only the essential idea.

Function of an Artificial Neuron (AN):

1. Receive a Input
2. Combine Feature Values of the Input
3. Decide based on the combination(s)
4. Output the Decision

Input

An input is an observation/example/data point in the feature space, denoted by X or (X, y) , where y is the target variable and $X = [x_i]$ is the feature values of the data point.

Combining the Feature Values of the Input

Every AN combines the feature values of the received input, somehow.

An standard AN combines, linearly, all feature values of the input data point together into a Linear **Weighted Sum**, using its parameters, namely, **Weights** and **Bias**. Then, this Weighted Sum is fed into an **Activation Function**, to decide the output of the AN.

Weights

Features, not necessarily, have equal influence on AN output. A trained AN has **Weights** parameter, denoted by $W = [w_i]$, tentatively learned during training by maintaining an aim of optimizing some objective function defined by the algorithm, with w_i being the **weight** or the **measure of influence** of feature x_i on AN output.

Bias

Linear combination of the feature values of an input requires another parameter called **Bias**, denoted by **b**, additional to **Weights**. This parameter too is tentatively learned during training, aimed at optimization of the objective function defined by the algorithm.

Weighted Sum Function, z

$$z \equiv z_{W,b}(X) \equiv W \cdot X + b$$

The operation denoted by the dot ($\cdot$) above is Scalar/Dot Multiplication from Elementary Vector Algebra.

$$W \cdot X = w_1x_1 + w_2x_2 + \dots + w_ix_i + \dots + w_nx_n$$

Yes, this is how a standard AN linearly combines the feature values of an input observation, using its Weights and Bias. z is the Linear Weighted Sum, which is the input to the Activation Function.

Activation Function, a

$$a \equiv a(z)$$

The activation function a is a function of z , the Linear Weighted Sum.

$a(z)$ can be a function of any kind— Linear or Non-Linear— Continuous or Discontinuous— Differentiable Non-differentiable etc.

And a is the AN output.

Without an activation function, the Linear Weighted Sum would be the output of the AN. But the Linear Weighted Sum is just a linear transformation on an input observation. Our AN would be just a linear model. Thus, our Neural Network, or DL Model, which is a stack of ANs, would be just linear in its capability. All the same with a linear activation function too.

Also, the following function was not invented by DL.

$$z \equiv z_{w,b}(X) \equiv W \cdot X + b$$

It's exactly the linear component used in Linear Regression, Logistic Regression, Perceptrons and Linear SVM.

DL is different from the Classical ML Model because of Activation Function. DL builds upon this linear transformation by stacking many such units together with nonlinear activation functions.